

Agent Dobby

Agentic AI-Powered Swing Trading Platform

CMU Executive Education · Agentic AI Program · Module 7 Capstone

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The Challenge

Manual swing trading is time-consuming, emotion-driven, and difficult to scale



Time-Intensive Research

Scanning news, charts, and indicators for dozens of stocks every single day



Cognitive Overload

Tracking 50+ technical indicators, patterns, and market signals simultaneously



Emotion-Driven Decisions

Fear and greed override systematic rules, leading to inconsistent outcomes



Inconsistent Execution

Manual order placement introduces delays, errors, and missed opportunities

40+

Hours/week on research

68%

Trades affected by emotion

1,000+

Technical signals to monitor

Meet Agent Dobby

Autonomous · Intelligent · Disciplined

Agent Dobby is a multi-agent AI system that autonomously researches, evaluates, and executes swing trades
— applying systematic rules, not gut feelings.



Researches

Ingests news via RAG pipeline
Builds semantic context on
every ticker being considered



Decides

Two-level Tree-of-Thought BFS
Filters by sentiment → then by
50+ technical conditions



Executes

Routes approved orders to Alpaca
Manages FIFO trade lifecycle
Monitors portfolio continuously

System Architecture

Five-layer pipeline from user chat to brokerage execution



DATA LAYER



Multi-Agent Design

Specialised agents with distinct responsibilities, coordinated by LangGraph state management



Supervisor

PortfolioManagerAgent

Orchestrates the full trade lifecycle — evaluates holdings, coordinates research and buy evaluation, routes execution orders



Analyst

ResearchAgent

Queries ChromaDB for relevant news summaries, applies RAG-based sentiment scoring against target tickers



Technician

BuyTechEvaluatorAgent

Evaluates 50+ technical conditions from the setup registry; applies Level-2 BFS filtering to candidate tickers



Risk Monitor

HoldingsEvaluatorAgent

Assesses current open positions in parallel; triggers sell or hold signals based on stop-loss and target conditions



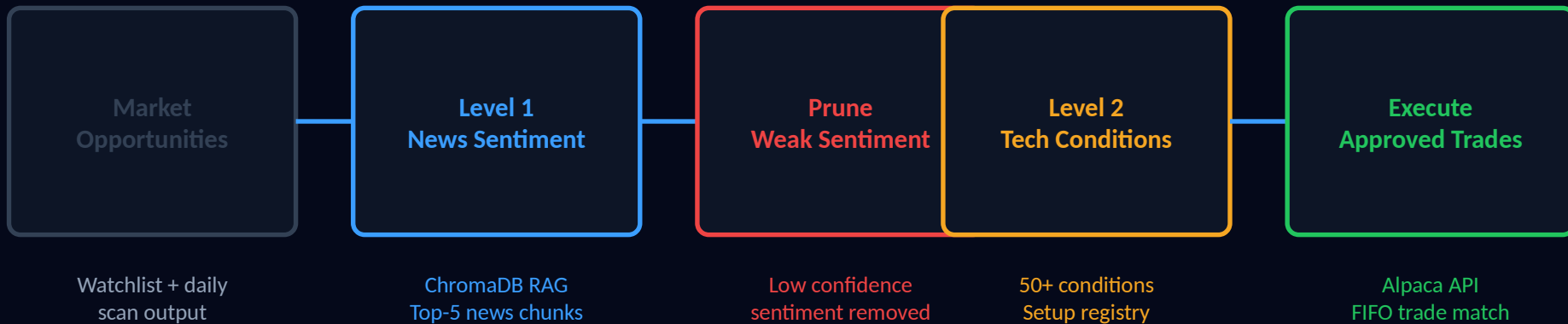
Interface

ChatAgent

LangGraph ReAct agent handling 18 action types — portfolio queries, backtest requests, watchlist management

Tree-of-Thought Decision Engine

Two-level BFS pipeline prunes candidate tickers before any trade is executed



BFS Pruning

Eliminates poor candidates early, reducing compute and risk



Two-Level

Qualitative news filter → quantitative technical filter



Setup Registry

50+ condition factories — extensible via JSON config

RAG-Powered News Intelligence

Semantic search connects live market sentiment to every trading decision



3×/day

Ingestion schedule
(7AM · 12PM · 8PM ET)

Top-5

Chunks retrieved
per semantic query

Local

ChromaDB — no cloud
dependency or cost

Haiku

Claude Haiku used
for cost-efficient summaries

Nine-Level Safety Guardrails

Safety-first architecture — multiple independent safeguards before any capital is deployed

01

Code Review Gate

All logic reviewed before deployment

02

Model Switching

Fallback models for API failures

03

Paper Trading Mode

Simulate before live execution

04

System Prompt Bounds

Agent scope constrained by design

05

Research Evaluation

Sentiment threshold before Level 2

06

Back-Tested Setups Only

Only statistically validated setups

07

Portfolio Constraints

Max position size and cash limits

08

Human-in-the-Loop

Planned approval gate for large trades

09

Monitoring & Recovery

APScheduler + failure detection

Technology Stack

Production-grade tools, fully open-source where possible

AI / ML

- LangGraph
- LangChain
- Claude (Anthropic)
- all-MiniLM-L6-v2
- ChromaDB

Backend

- FastAPI
- SQLAlchemy
- APScheduler
- Python 3.11+

Data

- PostgreSQL
- SQLite
- Alpha Vantage API
- Yahoo Finance RSS

Brokerage & UI

- Alpaca Markets API
- React + Vite
- pptxgenjs
- reportlab



Live Demo

Agent Dobby in action

What's Next

Planned enhancements to move Agent Dobby from local prototype to production platform



Human-in-the-Loop Approval

Planned notification + approval workflow for high-value trades before execution, enabling human oversight without breaking automation.



AWS Cloud Deployment

Containerise agents with Docker, deploy to ECS/Fargate, add RDS for PostgreSQL, and CloudWatch for monitoring and alerting.



Multi-Strategy Support

Extend beyond swing trading — scalping, momentum, and mean-reversion strategies configurable via the setup registry.



Portfolio Optimisation

Integrate Modern Portfolio Theory — dynamic position sizing, correlation-aware allocation, and drawdown-constrained rebalancing.